

Lecture 11

Infographics, Other Systems

[Data Visualization · 1-DAV-105](#)

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Infographics

Some examples of infographics

Several examples that are close to data visualization:

- Income by religious group in US ([image](#), [website](#))
- Deadliest pandemics ([image](#), [website](#))
- War casualties ([website](#))
- Game of Thrones relationships ([image](#), [website](#))

Some explain other types of information:

- Famous people with learning disabilities ([website](#))
- Preference polling in Slovakia ([website](#))

Data visualization (DV) vs infographics (IG)

- **Target audience:** IG general public, DV often experts
- **Storytelling:** often in IG, can be created from multiple DV
- **Design and aesthetics:** more elaborate in IG, includes graphics elements and clipart (considered chart junk in DV)
- **Process of creation:** many simple tools for DV, IG time consuming, often created by collaboration of data analysis, domain experts and graphic designers

<https://www.statsilk.com/blog/real-difference-between-infographics-and-data-visualizations>

Other visualization tools

Other visualization tools

Spreadsheets

- [Excel](#)
- [Google sheets](#)

Well-know commercial tools

- [Tableau](#): Advanced visualization tools
- [Flourish](#)
- [Microsoft Power BI](#): Interactive data visualization with a focus on business intelligence ([our university](#)); see elective course [Data Analysis in Power BI](#)

Other visualization tools

System R

- Programming language for statistical computing
- Together with Python, very popular in data science, built-in [plots](#)
- Colab has R runtime
- Library [ggplot2](#) based on Grammar of Graphics ([cheatsheet](#))

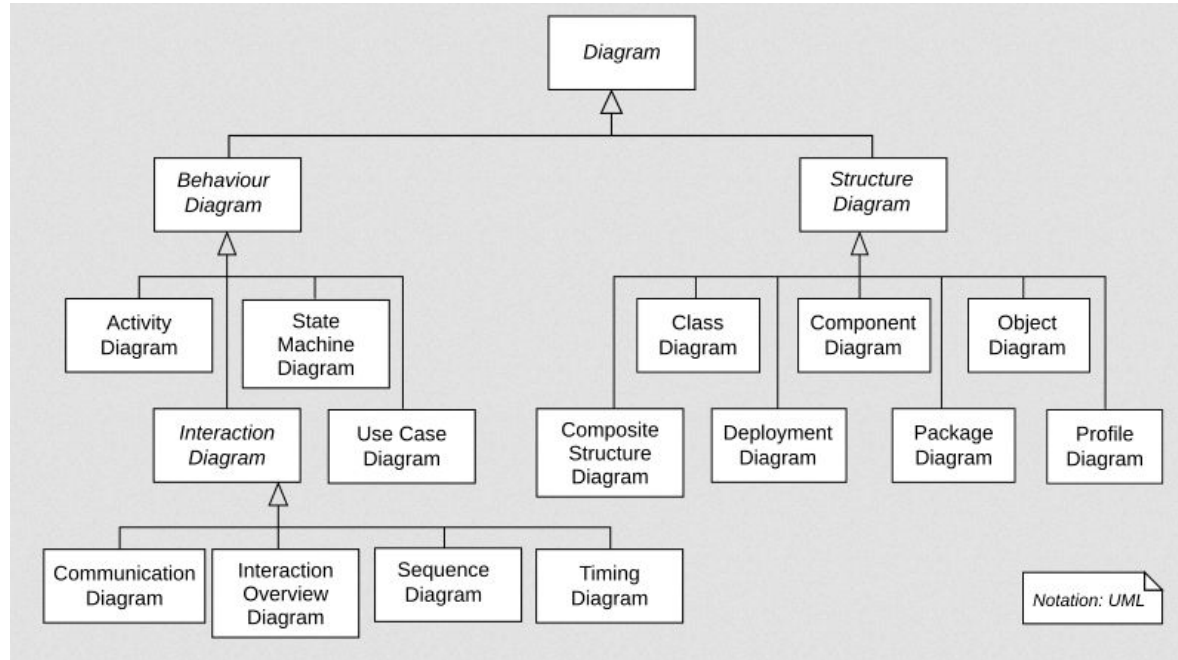
Javascript

- Programming language popular in web programming
- [Google charts](#) for Javascript (see [Data management](#) course)
- [D3.js](#) library (Data-Driven Documents)
- [Vega-Lite](#) uses javascript to [embed](#) plots specified as json

Specialized visualizations

UML diagrams in computer science

Display relationships between components and aspects of software



https://commons.wikimedia.org/wiki/File:UML_diagrams_overview.svg

Waterfall chart

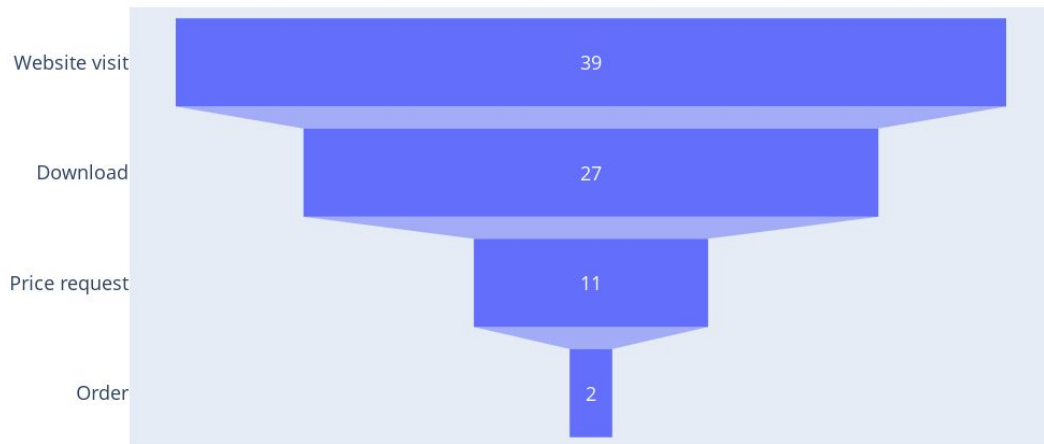
- Used in business analysis: financial, inventory, human resources etc.
- Displays effects decreasing or increasing a given value
- The first and last bars display starting and final value
- Intermediate columns float, displaying changes from previous total ([more info](#))



https://commons.wikimedia.org/wiki/File:Waterfallchart_ex2.jpg

Funnel charts

- Display losses within a business process, e.g. from website visit to actual purchase
- Horizontal bar chart with centered bars
- Different [funnel plot](#) in medical meta-analyses of multiple publications



Made in [plotly](#)

Candlestick chart

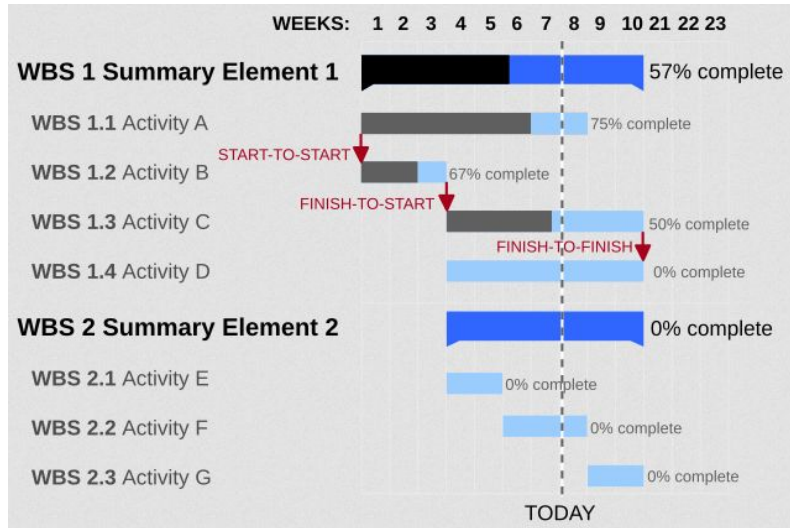
- Similar to boxplot, used in financial data, e.g. stocks, currency exchange rates
- Line: minimum and maximum, box: opening and close, color: increase or decrease

https://commons.wikimedia.org/wiki/File:Candlestick_Chart_in_MetaTrader_5.png



Gantt chart

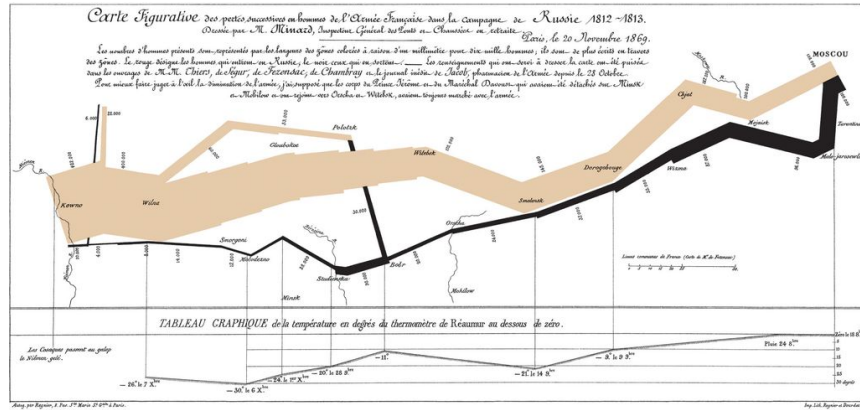
- Used in management to display project schedule with different tasks and their planned duration
- Can also display current status of tasks and their dependencies



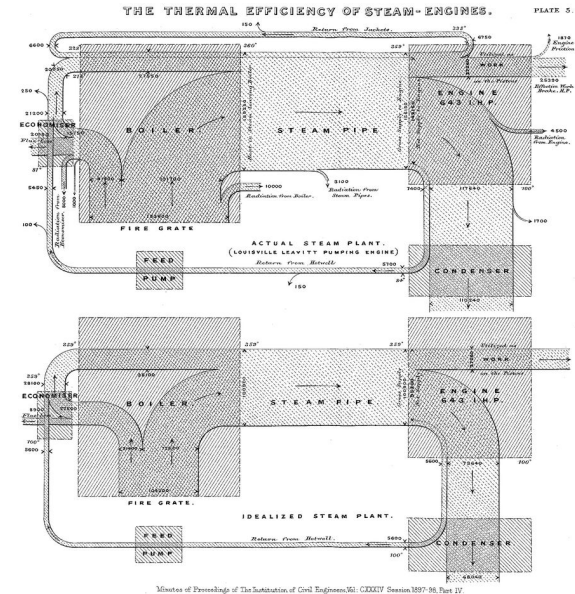
<https://commons.wikimedia.org/wiki/File:GanttChartAnatomy.svg>

Sankey diagram

- flows in a system, line width proportional to amount
- Minard's invasion of Russia 1869, Sankey's energy in steam engine 1898
- [Energy in EU](#)



<https://commons.wikimedia.org/wiki/File:Minard.png>



https://commons.wikimedia.org/wiki/File:JIE_Sankey_V5_Fig1.png

Another interesting book

Claus O. Wilke 2019 [Fundamentals of Data Visualization](#)

The author is a [scientist](#) (actually working bioinformatics)

The book is written in R, with [source code](#) available

We will look at several examples from this book

Forensic glass composition

- [Dataset](#)
- 214 glass samples
- Content of 7 elements (Na, Mg, Al, Si, K, Ca, Ba, Fe)
- Classification into 4 classes: window, headlamp, tableware, container
- Any ideas for visualization?
- What if we are interested in associations among variables?

Forensic glass composition

- [Dataset](#)
- 214 glass samples
- Content of 7 elements (% of mass)
- Classification into 4 classes: window, headlamp, tableware, container
- From the book:
 - [correlogram 1](#) ([image](#))
 - [correlogram 2](#) ([image](#))
 - [PCA](#) ([image](#)); [meaning of axes](#) ([image](#))
- How can we use PCA to learn something about a new sample?
- Notice that PCA plot uses both shape and color - what is the advantage?

CO2 emissions per person

- 166 countries
- Years 1970 and 2010 for each country (called paired data)
- How would you visualize?
- What questions can you ask?
- What if we are interested in only a small subset of countries?

CO2 emissions per person

- 166 countries
- Years 1970 and 2010 for each country (called paired data)
- From the book:
 - [scatterplot](#) ([image](#))
 - [slopegraph 1](#) ([image](#))
 - [slopegraph 2](#) ([image](#)) for years 2000, 2005, 2010
- Scatterplot of paired data has x-y diagonal
- Hard to judge distance from diagonal
- Mean-difference or sum-difference plot transforms coordinates ([example](#))

Comparing two time series

- Average temperature in different days of a year in two cities
- How would you plot this?

Comparing two time series

- Plot each against time in one linegraph
- Plot each against time in two graphs next to each other
- Scatterplot of one temperature against other
 - loses time information
 - can be added back as color
- Book uses [connected scatterplot](#) ([image](#))
 - another [example](#) ([image](#)); compared to [two line graphs](#) ([image](#))

Follow-up courses

- [Data Management](#) various techniques for processing data BIN, DAV 2S
- [Database Systems](#) BIN, DAV 3W
- [Fundamentals of Probability and Statistics](#) BIN 3W, DAV 2W
- [Statistical Methods](#) DAV 3W
- [Principles of Data Science](#) DAV 3W
- [Network Science](#) DAV 3S